



Computer Engineering Department

Course name: Web Technologies

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Topic: Report of LAB - 8

LAB 8: Write a XML file and display the content using XSL, Validate XML file using DTD Validate XML file using XS(XML Schema).

Consider the following XML document fragment.

The root element is "Employee_Info"

As the content for "Employee_Info," "Employee" occurs 0 or more times

As contents of "Employee," "Name," "Department," "Telephone," and "Email" elements occur once in respective order

"Name," "Department," "Telephone," and "Email" contents are text strings

"Employee" has an attribute called "Employee_Number"

"Employee_Number" content must be int type

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8.1 Brief introduction of XML, XSL, XSD, DTD::

1. XML (Extensible Markup Language)

- XML is used to store and transport data in a structured format.
- It does not have predefined tags like HTML; you define your own.

2. Namespaces in XML

- XML Namespaces help avoid element name conflicts when combining XML documents from different sources.
- Defined using the **xmlns** attribute.

3. XSL (Extensible Stylesheet Language)

- Used for transforming and styling XML data.
- XSLT (XSL Transformations) is commonly used to convert XML to other formats like HTML.

4. XSD (XML Schema Definition)

- Defines the structure, data types, and constraints for an XML document.
- Supports minOccurs, maxOccurs attributes to control element occurrences.
- minOccurs & maxOccurs in XSD
 - minOccurs="0" → Element is optional.
 - maxOccurs="1" → Element can appear at most once.
 - maxOccurs="unbounded" → Element can appear any number of times.

5. DTD (Document Type Definition)

- Defines the basic structure of an XML document.
- It uses #PCDATA (Parsed Character Data) and #CDATA (Character Data).

6. #PCDATA (Parsed Character Data)

- Parsed by the XML processor.
- Can include **text and entity references**.

7. #CDATA (Character Data)

- Used to include unparsed text.
- The XML processor does not interpret special characters inside CDATA.

8.2 Exercise 1: Write an xml (EMP.XML) File for above description.

XML:

```
<?xml version="1.0" encoding="UTF-8"?>
```

```
<?xml-stylesheet type="text/xsl" href="style.xsl"?>
```

```
<employee_info xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:noNamespaceSchemaLocation="emp.xsd">
```

```
  <employee employee_number="1">
```

```
    <Name>Amit</Name>
```

```
    <Department>CP</Department>
```

```
    <telephone>9879879879</telephone>
```

```
    <email>abc@gmail.com</email>
```

```
  </employee>
```

```
  <employee employee_number="2">
```

```
    <Name>Rahul</Name>
```

```
    <Department>IT</Department>
```

```
    <telephone>9876543210</telephone>
```

```
    <email>rahul@example.com</email>
```

```
  </employee>
```

```
</employee_info>
```

8.3 Exercise 2: Write a document type definition (EMP.DTD) file to validate above XML file

DTD:

```
<!ELEMENT employee_info (employee*)>  
<!ELEMENT employee (Name,Department,telephone,email) >  
<!ATTLIST employee employee_number CDATA #REQUIRED>  
<!ELEMENT Name (#PCDATA)>  
<!ELEMENT Department (#PCDATA)>  
<!ELEMENT telephone (#PCDATA)>  
<!ELEMENT email (#PCDATA)>
```

8.4 Exercise 3: Write an xml schema EMP.xsd file to validate above XML file.

XSD:

```
<?xml version="1.0" encoding="UTF-8"?>

<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="employee_info">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="employee" maxOccurs="unbounded">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="Name" type="xs:string"/></xs:element>
              <xs:element name="Department" type="xs:string"/></xs:element>
              <xs:element name="telephone" type="xs:string"/></xs:element>
              <xs:element name="email" type="xs:string"/></xs:element>
            </xs:sequence>
            <xs:attribute name="employee_number" type="xs:int"/></xs:attribute>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

8.5 Exercise 4: Write an xml style sheet file (EMP.xsl) file to print above XML data in table format.

XSL:

```
<?xml version="1.0" encoding="UTF-8"?>
<xsl:stylesheet version="1.0" xmlns:xsl="http://www.w3.org/1999/XSL/Transform">

  <xsl:template match="/">

    <html>

      <head>

        <title>Employee Information</title>

        <style>

          table {border-collapse: collapse; width: 50%;}

          th, td {border: 1px solid black; padding: 10px; text-align: left;}

          th {background-color: lightgray;}

        </style>

      </head>

      <body>

        <h2>Employee Information</h2>

        <table>

          <tr>

            <th>Name</th>

            <th>Department</th>

            <th>Telephone</th>

            <th>Email</th>

          </tr>

          <xsl:for-each select="employee_info/employee">
```

```
<tr>
  <td><xsl:value-of select="Name"/></td>
  <td><xsl:value-of select="Department"/></td>
  <td><xsl:value-of select="telephone"/></td>
  <td><xsl:value-of select="email"/></td>
</tr>
</xsl:for-each>

</table>
</body>
</html>
</xsl:template>

</xsl:stylesheet>
```

8.6 OUTPUT:

Employee Information

Name	Department	Telephone	Email
Amit	CP	9879879879	abc@gmail.com
Rahul	IT	9876543210	rahul@example.com
